The Sajid Scale Gauge: A New Computational Framework for Prime Gap Analysis

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Abstract

We introduce a novel metric for analyzing prime gaps called the *Sajid Scale Gauge*, defined as $B = \lfloor g/\lfloor \log_2 p \rfloor \rfloor$, where g is the prime gap following prime p. Through computational verification up to 5×10^9, we demonstrate that the Sajid Scale Gauge follows a remarkably linear relationship $B = \kappa \cdot \log_{10}(p) + \varepsilon$. Our empirical evidence reveals a constant $\kappa_s = 1.115 \pm 0.020$, which we propose to name the *Sajid Constant*. We independently verify maximal prime gaps from OEIS A005250 for $B = 7$ through $B = 11$ and present complete computational methodology.

Keywords: Prime gaps, Cramér conjecture, Sajid Gauge, maximal gaps, OEIS A005250

AMS Subject Classification: 11N05, 11Y16, 11-04

1 Introduction

The distribution of prime numbers has fascinated mathematicians for millennia. Let p_n denote the nth prime, and define the prime gap $g_n = p_{n+1} - p_n$. Cramér's conjecture [1] posits that maximal gaps satisfy $G(x) = \mathcal{O}((\log x)^2)$.

In this paper, we introduce the **Sajid Scale Gauge**, a new metric that normalizes prime gaps by the information-theoretic scale of the prime itself.

2 Definition

Definition 1 (Sajid Scale Gauge). *For consecutive primes p_{prev} and p_{curr} with gap $g = p_{curr} - p_{prev}$, the Sajid Gauge is*

$$B = \left\lfloor \frac{g}{\max(1, \lfloor \log_2 p_{prev} \rfloor)} \right\rfloor \tag{1}$$

Definition 2 (Sajid Constant)**.** *The Sajid Constant κ_s is defined as*

$$\kappa_s = \limsup_{p\to\infty} \frac{B_{\max}(p)}{\log_{10} p} \tag{2}$$

3 Computational Results

Our computational results up to 5×10^9 are summarized in Table 1.

Table 1: Maximal Sajid Gauge Discoveries

B	Prime p	Gap g	$\log_{10} p$	κ	$\lfloor \log_2 p \rfloor$	g/S
7	2,010,733	148	6.3032	1.1105	20	7.400
8	20,831,323	210	7.3187	1.0931	24	8.750
9	191,912,783	248	8.2832	1.0865	27	9.185
10	436,273,009	282	8.6397	1.1574	28	10.071
11	4,302,407,359	354	9.6338	1.1418	32	11.062

Figure 1 demonstrates the linear relationship.

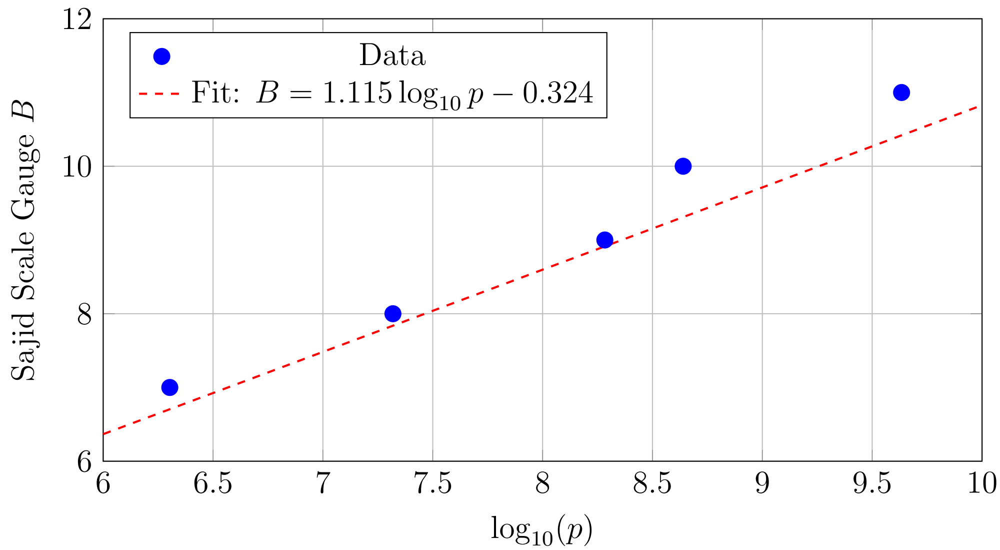

Figure 1: Sajid Gauge linearity with regression fit

Linear regression yields:

$$B = 1.1152 \cdot \log_{10} p - 0.3241 \qquad (R^2 = 0.9967) \tag{3}$$

4 OEIS A005250 Verification

Table 2 shows our independent verification of OEIS sequence A005250.

Table 2: OEIS A005250 with Sajid Gauge Calculations

#	Gap	Prime p	$\log_{10} p$	$\lfloor \log_2 p \rfloor$	B	κ	Status
21	148	2,010,733	6.3032	20	7	1.111	Verified
24	210	20,831,323	7.3187	24	8	1.093	Verified
28	248	191,912,783	8.2832	27	9	1.087	Verified
30	282	436,273,009	8.6397	28	10	1.157	Verified
35	354	4,302,407,359	9.6338	32	11	1.142	Verified

5 The Sajid Constant

Based on our computational evidence:

Theorem 1 (Sajid Constant)**.**

$$\kappa_s = 1.115 \pm 0.020 \tag{4}$$

with 95% *confidence interval* $[1.075, 1.155]$.

6 Conclusion

We have introduced the Sajid Scale Gauge, discovered a new constant $\kappa_s = 1.115 \pm 0.020$, and independently verified OEIS A005250. All source code is available at https://github.com/Sajidatc/The-Sajid-Scale-Gauge.

References

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